# From Data Discovery to Analytics-Ready Data: A Survey of Search, Preparation, and Integration in Data Lakes

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## 1. Description

Data lake pipelines are commonly described as three phases: *discovery & search* (finding relevant tables), *preparation & wrangling* (cleaning and structuring data for analysis), and *integration & matching* (aligning schemas and reconciling records). This survey will review recent work across all three phases and organize the literature by the *artifacts each phase produces* and the *assumptions that these artifacts encode*.

In the literature, evaluation is often reported using *phase-local* metrics that are not directly comparable across phases. A key goal of this survey will be to summarize how the papers define phase outputs, what datasets and benchmarks they use, and which evaluation setups are most useful for analytics-ready data.

## 2. Method

I will conduct a structured literature review consisting of:

- a reproducible paper collection protocol (venues/years, search strings, inclusion/exclusion criteria);
- a taxonomy that maps papers to phase outputs and problem settings; and
- a synthesis that compares the assumptions of the problem, the artifacts produced, and the evaluation practices between phases.

Key references include LakeBench [2] (benchmarks for discovery), ALITE [3] (table integration semantics) and HAIPipe [1] (human–machine preparation pipelines).

## 3. Plans and Timeline

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| **Weeks 1–2** | Finalize scope (years, venues, and tasks). Write the review protocol: search strings and inclusion/exclusion criteria. Build an initial reading list from benchmark and system anchor works. |
| **Weeks 3–4** | Screen and expand the corpus (including backward/forward citation chasing). Assign papers to a first-pass taxonomy for discovery and search, preparation and wrangling, and integration and matching. |
| **Weeks 5–6** | Extract evaluation details (datasets, labels) using a consistent coding template. |
| **Weeks 7** | Draft the survey synthesis by phase, emphasizing phase outputs and how papers define inputs/outputs, assumptions, and evaluation. |
| **Weeks 8** | Finalize the write-up: taxonomy figure(s), comparison table(s) for datasets, and a concise set of open problems. |

## References

[1] S. Chen, N. Tang, J. Fan, X. Yan, C. Chai, G. Li, and X. Du. Haipipe: Combining human-generated and machine-generated pipelines for data preparation. *Proc. ACM Manag. Data*, 1(1), May 2023.

[2] Y. Deng, C. Chai, L. Cao, Q. Yuan, S. Chen, Y. Yu, Z. Sun, J. Wang, J. Li, Z. Cao, K. Jin, C. Zhang, Y. Jiang, Y. Zhang, Y. Wang, Y. Yuan, G. Wang, and N. Tang. Lakebench: A benchmark for discovering joinable and unionable tables in data lakes. *Proc. VLDB Endow.*, 17(8):1925–1938, Apr. 2024.

[3] A. Khatiwada, R. Shraga, W. Gatterbauer, and R. J. Miller. Integrating data lake tables. *Proc. VLDB Endow.*, 16(4):932–945, Dec. 2022.